

Fin404 Derivatives

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Dividend Derivatives

The goal of this project is to study the construction of the volatility index (VIX) and to analyze different approaches to the modelling of volatility.

Part 1. Documentation

Provide a basic presentation (maximum 3 pages) of the market for dividend-related derivatives taking as reference the contracts traded on EUREX. This part of your report is meant as an introductory briefing. As such it should not be technical, but it can include basic equations, numbers, and graphics as necessary. You are free to structure and organize your presentation as you see fit. However, your report should at the minimum address the following points:

- History of the market for dividend-related derivatives
- What are the main dividend-related derivatives traded on EUREX? How do they function? What are possible use cases for these different contracts?
- Why do dividend futures exist? What do they allow investors to achieve?
- What is the equity term structure? How does it relate to dividend futures?

Part 2. Model-free pricing

Consider an index with spot price S_t and assume that European puts and calls on this index with *any* maturity T and *any* strike k are traded. Assume further that the riskfree

rate r is constant and denote by $\mathbb{Q}$ an arbitrary EMM.

1. Show that the market value at time t of the dividends paid by the index in the time interval $[t, T]$ satisfies

$$\mathcal{D}_t(T) = S_t - B_t(T)k - C_t(T, k) + P_t(T, k), \quad \forall k,$$

where $B_t(T)$ and $C/P_t(T, k)$ denote the prices of a zero coupon with maturity T and a European call/put on the index with maturity T and strike k .

2. Show that

$$\mathcal{D}_t(T) = E_t^{\mathbb{Q}} \left[\int_t^T e^{-r(s-t)} dD_s \right]$$

for any finite horizon T where D_t denotes the (nondecreasing) cumulative dividend process of the index.

3. Fix arbitrary $t \leq T_0 < T_1$. Construct at date t a portfolio of the index, calls, puts, and zero-coupons whose value at date T_0 equals $\mathcal{D}_{T_0}(T_1)$.
4. Construct at date t a portfolio of the index and infinitely many calls, puts, and zero coupons whose value at date T_0 equals

$$\mathcal{D}_{T_0}^*(T_1) := E_{T_0}^{\mathbb{Q}} \left[\int_{T_0}^{T_1} dD_s \right] = E_{T_0}^{\mathbb{Q}} [D_{T_1} - D_{T_0}]$$

Show that the $[T_0, T_1]$ -dividend futures price at time $s \leq T_0$ can be expressed as $e^{r(T_0-s)}X_s$ where X_s denotes the value of the portfolio you just constructed.

Hint. Write

$$\int_{T_0}^{T_1} dD_s = \int_{T_0}^{T_1} e^{r(s-T_0)} e^{-r(s-T_0)} dD_s$$

and combine integration-by-parts with Fubini's theorem to derive a decomposition of the payoff $\mathcal{D}_{T_0}^*(T_1)$ in terms of $\{\mathcal{D}_{T_0}(s) : T_0 \leq s \leq T_1\}$.

5. What is the $[T_0, T_1]$ -dividend futures price at time $s \in (T_0, T_1]$?
6. What are the practical limitations of the above model-free analysis?

Part 3. Model-based pricing

1. Why is it important to use a model that simultaneously prices index derivatives and dividend-related index derivatives?

Assume from now on that the index price S_t , its dividend rate C_t , and its cumulative dividend process D_t jointly evolve according to

$$\begin{aligned}dS_t &= (rS_t - C_t) dt + \sigma \sqrt{C_t} dB_t^{\mathbb{Q}} \\dC_t &= \lambda (C^* - C_t) dt + \phi \sqrt{C_t} dB_t^{\mathbb{Q}} \\dD_t &= C_t dt\end{aligned}$$

where B_t is a scalar $\mathbb{Q}$ -Brownian motion and $(\sigma, \phi, r, \lambda, C^*)$ are constant parameters such that

$$r, \lambda, C^* > 0 \text{ and } \phi^2 < 2\lambda C^*.$$

You may use without proof the fact that under these conditions (1) admits a unique strong solution such that $S_t, C_t, D_t > 0$ at all times.

2. Show that

$$E_t^{\mathbb{Q}}[C_s] = C^* + e^{-\lambda(s-t)} (C_t - C^*), \quad 0 \leq t \leq s < \infty,$$

and use this property to establish that

$$\int_0^t e^{-rs} \sqrt{C_s} dB_s^{\mathbb{Q}}$$

and

$$e^{-rt} S_t + \int_0^t e^{-rs} C_s ds$$

are both *true* martingales under $\mathbb{Q}$.

3. Show that the index price satisfies

$$F_t := E_t^{\mathbb{Q}} \left[\int_t^{\infty} e^{-r(s-t)} C_s ds \right] = a + b (C_t - C^*)$$

for some constants (a, b) to be determined. What does this process represent?

4. Show that

$$S_t = E_t^{\mathbb{Q}} \left[e^{-r(T-t)} S_T + \int_t^T e^{-r(s-t)} C_s ds \right], \quad 0 \leq t \leq T < \infty.$$

Combine this property with the result of the previous question to prove that the following properties

- a. $\sigma = \frac{\phi}{r+\lambda}$ and $S_0 = F_0$
- b. $S_t = F_t$ for all $t \geq 0$
- c. $\lim_{T \rightarrow \infty} E_t^{\mathbb{Q}} [e^{-r(T-t)} S_T] = 0$ for all $t \geq 0$

are equivalent.

- 5. Assume from now on that the above equivalent conditions are satisfied. Show that the dividend yield $\delta_t := C_t/S_t$ is bounded and evolves according to an *autonomous* stochastic differential equation.
- 6. Provide an expression for the futures price of the index at date $t \in [0, T]$ for delivery at date T as a function of (t, S_t, C_t)
- 7. Provide an expression for the $[T_0, T_1]$ -dividend futures price at date $t \in [0, T_1]$ as a function of (t, S_t, C_t) and the cumulative dividend $D_{t \vee T_0} - D_{T_0}$ paid by the index between dates T_0 and $t \vee T_0 = \max\{t, T_0\}$

8. Show that

$$\begin{aligned} E_t^{\mathbb{Q}} [e^{-wC_{T_1}}] &= e^{-a_0(T_1-t) - b_0(T_1-t)C_t} \\ E_t^{\mathbb{Q}} [e^{-w(D_{T_1} - D_{T_0})}] &= e^{-a(t) - b(t)C_t - w(D_{t \vee T_0} - D_{T_0})} \end{aligned}$$

for some deterministic functions (a_0, b_0, a, b) to be determined. Make sure to note how these functions depend on the exponent $w \geq 0$.

9. Let

$$P_t(k) := e^{-r(T_1-t)} E_t^{\mathbb{Q}} [(k - D_{T_1} + D_{T_0})^+]$$

denote the price at time $t \leq T_1$ of a dividend put with maturity T_1 , reference interval

$[T_0, T_1]$, and strike k . Use Fubini's theorem to show that

$$\widehat{P}_t(w) := \int_0^\infty e^{-wk} P_t(k) dk = \frac{1}{w^2} e^{-\widehat{a}(t) - \widehat{b}(t) C_t - w(D_{t \vee T_0} - D_{T_0})}$$

for all $w > 0$ and some deterministic functions $(\widehat{a}, \widehat{b})$ to be determined. Propose a numerical method that allows to inverse this Laplace transform to obtain the price of the dividend put option at any given strike.

10. Show that

$$E_t^{\mathbb{Q}} \left[(D_{T_1} - D_{T_0})^n \right] = (-1)^n \frac{d^n}{dw^n} E_t^{\mathbb{Q}} \left[e^{-w(D_{T_1} - D_{T_0})} \right] \Big|_{w=0}.$$

Use this result to derive an explicit expression for the moment on the left as a function of (t, C_t) and the (unevaluated) derivatives

$$\left\{ \frac{d^n a(\cdot)}{dw^n}, \frac{d^n b(\cdot)}{dw^n} \right\} \Big|_{w=0}$$

of the functions you derived in the previous question and provide an explicit formula for the futures price associated with a contract that settles at date T_1 on the *square* of the cumulative dividends paid in $[T_0, T_1]$.

11. Use the fact that

$$\frac{1}{dx} \mathbb{Q}[C_T \in dx] = a e^{-a(x + e^{-\lambda T} C_0)} \left(e^{\lambda T} \frac{x}{C_0} \right)^{\frac{q}{2}} I_q \left(2a \sqrt{e^{-\lambda T} x C_0} \right)$$

where

$$q = \frac{2\lambda C^*}{\phi^2} - 1$$

$$a = a(T) := \frac{2\lambda}{\phi^2 (1 - e^{-\lambda T})}$$

and I_q is the modified Bessel function of the first kind (see [here](#)) to derive an explicit expression for the density of the index

$$\frac{1}{ds} \mathbb{Q}[S_T \in ds]$$

as a function of T and the initial price of the index. Give an expression for the initial price of an index call as an (unevaluated) integral against that density.

Lag k	Futures price $f_0^{0.25+[k,k+1]}$
0	17.1981
1	18.7223
2	19.4360
3	19.7445
4	19.8833
5	19.9623
6	19.9703
7	20.0066
8	20.0019
9	20.0031

Table 1: Dividend futures prices on October 1, 2025

12. Assume that today is October 1, 2025, that $r = 0.05$, and that the Euro Stoxx 50 is currently worth $S_0 = 394.118$. To calibrate the parameters $(\lambda, C^*, C_0, \phi)$ we will try to ensure that the model correctly prices:

- The Euro Stoxx 50 index
- Euro Stoxx dividend futures price with reference intervals

$$[0.25 + k, 0.25 + 1 + k], \quad k = 0, 1, \dots, 9$$

- An ATM call on the Euro Stoxx 50 with maturity 3M and Black-Scholes implied volatility $\hat{\sigma} = 0.04635$

Formulate the calibration problem as the minimization of the (unweighted) sum of squared deviations between market and model-implied prices. Solve the problem numerically for the optimal parameter values. Discuss the results you obtain and explain the main steps of your code in the report.

13. Suppose you are trying to replicate a 1Y ATM call on the Euro Stoxx 50 by investing in the riskless asset and a 2Y futures on the Euro Stoxx 50. Use your calibrated parameters to compute the initial number of futures contracts that your replicating portfolio should include.

General guidelines

1. The project is to be done in groups of at least 3 and no more than 4 students without communications between the groups.
2. The name and sciper number of each member of the group should appear clearly on the final report.
3. The final report should be typeset on no more than 15 A4 pages in 12pt font with 1.5cm margins and standard linespacing.
4. The code should be ready to run in one or more standalone files. If there are multiple files please also provide a readme with clear instructions.
5. The code should be commented with explanations for each line/block preferably in the code itself.
6. The code should not use any external financial engineering libraries or functions. Please develop your own. It may however include built-in optimization, differential equation solving, and root finding functions.
7. Your code and the final report should be uploaded on moodle.
8. Only one final report and one code archive per group can be uploaded. No need to also send your work by email.
9. Make sure to not simply copy existing material and to cite your sources. Your work will automatically be scanned by an antiplagiarism software.